

# Topology

沈威宇

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# 1 Topology (拓樸學)

## I Topological space (拓樸空間)

A topological space consists of a set  $X$  and a topology  $\mathcal{T}$  on it, denoted as  $(X, \mathcal{T})$ . Where the set  $X$  is the set of points in the space, and the topology  $\mathcal{T}$  is the set of subsets of  $X$  that satisfies:

1.  $\emptyset, X \in \mathcal{T}$
2. Closed under arbitrary unions:  $\forall S \subseteq \mathcal{T} : \bigcup_{A \in S} A \in \mathcal{T}$
3. Closed under finite intersection:  $\forall S \subseteq \mathcal{T}$  s.t.  $|S| < \infty : \bigcap_{A \in S} A \in \mathcal{T}$

## II Homeomorphism (同胚) or isomorphism (同構)

Topological spaces  $(X, \mathcal{T}_X)$  and  $(Y, \mathcal{T}_Y)$  are called homeomorphic if there exists a mapping  $f : X \rightarrow Y$  between them such that  $f$  is bijective and continuous and  $f^{-1}$  is continuous, written as  $X \cong Y$ , and  $f$  is called a homeomorphism between them.

## III Open set (開集)

In a topological space  $(X, \mathcal{T})$ , an open subset  $O \subseteq X$  is defined to be

$$O \in \mathcal{T}.$$

## IV Open neighborhood (開鄰域)

An open neighborhood of a point  $P$  in a topological space  $X$  is any open subset  $O \subseteq X$  such that  $P \in O$ .

An open neighborhood of a subset  $V$  in a topological space  $X$  is any open subset  $O \subseteq X$  such that  $V \subseteq O$ .

## V Interior (內部) of subset

The interior of a subset  $S$  of a topological space  $X$ , denoted by  $\text{int}_X S$ ,  $\text{int } S$ ,  $\text{int}_X(S)$ ,  $\text{int}(S)$ , or  $S^\circ$ , is defined as

$$\text{int}(S) = \{x \in S \mid \exists \text{ open } U \subseteq X \text{ s.t. } x \in U \subseteq S\}.$$

## VI Neighborhood (鄰域)

A subset  $U$  is a neighborhood of a point  $P$  if and only if there exists an open set  $O$  such that  $P \in O \subseteq U$ .

A subset  $U$  is a neighborhood of a subset  $V$  if and only if  $V \subseteq \text{int}(U)$ .

## VII Subspace topology

The subspace topology of a topological space  $(X, \mathcal{T})$  in a subset  $I$  of  $X$  is defined as the set

$$\mathcal{I} = \{U \cap I \mid U \in \mathcal{T}\}.$$

## VIII Topological embedding

A topological embedding is a homeomorphism onto its image with subspace topology.

## IX Limit point (極限點), cluster point, or accumulation point (集積點)

Let  $S$  be a subset of a topological space  $X$ . A point  $x$  in  $X$  is a limit point, cluster point, or accumulation point of the set  $S$  if every neighborhood of  $x$  contains at least one point of  $S$  different from  $x$  itself.

## X Closure (閉包)

The closure of a subset  $S$  of points in a topological space consists of all points in  $S$  together with all limit points of  $S$ .

## XI Closed set (閉集)

A subset  $A$  of a topological space  $(X, \mathcal{T})$  is called closed if its complement  $X \setminus A \in \mathcal{T}$ .

## XII Borel set (博雷爾集)

A Borel set  $B$  is any set in a topological space  $(X, \mathcal{T})$  that can be formed from open sets through the operations of countable union, countable intersection, and relative complement, that is,

1. Every element of  $\mathcal{T}$  is a Borel set.
2. For any indexed family  $(B_i)_{i=1}^{\infty}$  of Borel sets,  $\bigcap_{i=1}^{\infty} B_i$  is a Borel set.
3. If  $B, C$  are Borel sets, then  $B \setminus C$  is a Borel set.
4. Any other set is not a Borel set.

## XIII Filter

A filter on a set  $X$  is a family  $\mathcal{B}$  of subsets of  $X$  such that:

1.  $X \in \mathcal{B}$ .
2.  $\emptyset \notin \mathcal{B}$ .
3.  $A \in \mathcal{B} \wedge B \in \mathcal{B} \implies A \cap B \in \mathcal{B}$ .
4.  $A \subseteq B \subseteq X \wedge A \in \mathcal{A} \in \mathcal{B} \implies B \in \mathcal{B}$ .

## **XIV Base or basis (基)**

Given a topological space  $(X, \mathcal{T})$ , a base (or basis) for the topology  $\mathcal{T}$  (also called a base for  $X$  if the topology is understood) is a family  $\mathcal{B} \subseteq \mathcal{T}$  of open sets such that every open set of the topology can be represented as the union of some subfamily of  $\mathcal{B}$ .

The topology generated by a base  $\mathcal{B}$ , generally denoted by  $\tau(\mathcal{B})$  can be defined to be follows: A subset  $O \subseteq X$  is to be declared as open, if for all  $x \in O$ , there exists some  $B \in \mathcal{B}$  such that  $x \in B \subseteq O$ .

## **XV Prefilter or filter base**

$\mathcal{B}$  is called a prefilter if its upward closure  $\uparrow \mathcal{B}$  is a filter.

## **XVI Quotient space (商空間)**

Let  $X$  be a topological space and  $\cong$  be an equivalence relation on it. The quotient space  $Y$  of  $X$  by  $\cong$  is the quotient set of  $X$  by  $\cong$  equipped with the quotient topology, defined as  $U \subseteq Y$  is open in the quotient topology on  $Y$  if and only if  $\{x \in X \mid [x] \in U\}$  is open in  $X$  and  $V \subseteq Y$  is closed in the quotient topology on  $Y$  if and only if  $\{x \in X \mid [x] \in V\}$  is closed in  $X$ .

## **XVII Type of topological spaces**

### **i Distinct**

Point  $x$  and point  $y$  in a topological space are distinct if and only if they are not the same point.

### **ii Kolmogorov space (柯爾莫果洛夫空間) or $T_0$ space**

Two points in a topological space are topological distinguishable if and only if there exists an open set that contains one of them but not the other.

A topological space is called a Kolmogorov space or  $T_0$  space if and only if any two distinct points in it are topological distinguishable. This condition is called the  $T_0$  condition or the first separation theorem.

### **iii Kolmogorov quotient (space)**

The Kolmogorov quotient (space) of a topological space is the quotient space of it under the equivalence relation topological indistinguishability.

### **iv Fréchet space (弗蘭歇空間), accessible space, or $T_1$ space**

Two points  $x$  and  $y$  in a topological space are separated if and only if there exists open sets  $U, V$  such that  $x \in U \wedge y \notin U \wedge y \in V \wedge x \notin V$ .

A topological space is called a Fréchet space, accessible space, or  $T_1$  space if and only if any two distinct points in it are separated. This condition is called the  $T_1$  condition or the second separation theorem. A topological space is a  $T_1$  space implies it is a  $T_0$  space.



## **v Symmetric space or $R_0$ space**

A topological space is called a symmetric space or  $R_0$  space if and only if any two topological distinguishable points in it are separated.

## **vi Hausdorff space (郝斯多夫空間), separated space (分離空間), or $T_2$ space**

Points  $x$  and  $y$  in a topological space are separated by neighborhoods if and only if there exists a neighborhood  $U$  of  $x$  and a neighborhood  $V$  of  $y$  such that  $U$  and  $V$  are disjointed.

A topological space is called a Hausdorff space, separated space, or  $T_2$  space if and only if any two distinct points in it are separated by neighborhoods. This condition is called the  $T_2$  condition or the third separation theorem. A topological space is a  $T_2$  space implies it is a  $T_1$  space.

## **vii Urysohn space or $T_{2\frac{1}{2}}$ space**

Points  $x$  and  $y$  in a topological space are separated by closed neighborhoods if and only if there exists a closed neighborhood  $U$  of  $x$  and a closed neighborhood  $V$  of  $y$  such that  $U$  and  $V$  are disjointed.

A topological space is called a Urysohn space or  $T_{2\frac{1}{2}}$  space if and only if any two distinct points in it are separated by closed neighborhoods. A topological space is a  $T_{2\frac{1}{2}}$  space implies it is a  $T_2$  space.

## **viii Completely Hausdorff space, functionally Hausdorff space, or completely $T_2$ space**

Points  $x$  and  $y$  in a topological space  $X$  are separated by a (continuous) function if and only if there exists a continuous function  $f : X \rightarrow [0, 1]$  such that  $f(x) = 0$  and  $f(y) = 1$ .

A topological space is called a completely Hausdorff space, functionally Hausdorff space, or completely  $T_2$  space if and only if any two distinct points in it are separated by a continuous function. A topological space is a completely  $T_2$  space implies it is a  $T_{2\frac{1}{2}}$  space.

## **ix Regular space (正則空間)**

A topological space  $X$  is called a regular space if and only if for any closed set  $F$  and any point  $x \notin F$ , there exists a neighbourhood  $U$  of  $x$  and a neighbourhood  $V$  of  $F$  such that  $U$  and  $V$  are disjointed.

## **x Regular Hausdorff space or $T_3$ space**

A topological space is called a regular Hausdorff space or  $T_3$  space if and only if it is a regular space and a Hausdorff space.

## **xi Completely regular space**

A closed set  $F$  and a point  $x \notin F$  in a topological space  $X$  are separated by a (continuous) function if and only if there exists a continuous function  $f : X \rightarrow [0, 1]$  such that  $f(x) = 0$  and

$$F \subseteq f^{-1}(\{1\}).$$

A topological space  $X$  is called a completely regular space if and only if any closed set  $F$  and any point  $x \notin F$  in it are separated by a (continuous) function. A topological space is a completely regular space implies it is a regular space.

## **xii Completely regular Hausdorff space, Tychonoff space, completely $T_3$ space, or $T_{3\frac{1}{2}}$ space**

A topological space  $X$  is called a completely regular Hausdorff space, Tychonoff space, completely  $T_3$  space, or  $T_{3\frac{1}{2}}$  space if and only if it is a completely regular space and a Hausdorff space. A topological space is a completely  $T_3$  space implies it is a completely  $T_2$  space.

## **xiii Normal space (正規空間)**

Disjoint closed sets  $E$  and  $F$  in a topological space are separated by neighborhoods if and only if there exists a neighborhood  $U$  of  $E$  and a neighborhood  $V$  of  $F$  such that  $U$  and  $V$  are disjointed.

A topological space is called a normal space if and only if any two distinct closed sets in it are separated by neighborhoods. A topological space is a normal space implies it is a regular space.

## **xiv Urysohn's lemma**

Two closed sets  $E$  and  $F$  in a topological space  $X$  are separated by a (continuous) function if and only if there exists a continuous function  $f : X \rightarrow [0, 1]$  such that  $E \subseteq f^{-1}(\{0\})$  and  $F \subseteq f^{-1}(\{1\})$ .

The Urysohn's lemma states that, a topological space is a normal space if and only if any two disjoint closed sets in it are separated by a (continuous) function.

## **xv Normal Hausdorff space or $T_4$ space**

A topological space is called a normal Hausdorff space or  $T_4$  space if and only if it is a normal space and a  $T_1$  space. A topological space is a normal Hausdorff space implies it is a regular Hausdorff space.

## **xvi Completely normal space or hereditarily normal space**

A topological space is called a completely normal space or hereditarily normal space if and only if any subspace of it is a normal space.

## **xvii Completely normal Hausdorff space, $T_5$ space, or completely $T_4$ space**

A topological space is called a completely normal Hausdorff space,  $T_5$  space, or completely  $T_4$  space if and only if it is a completely normal space and a  $T_1$  space. A topological space is a completely normal Hausdorff space implies it is a normal Hausdorff space and a completely regular Hausdorff space.

### **xviii Perfectly normal space**

Two closed sets  $E$  and  $F$  in a topological space  $X$  are precisely separated by a (continuous) function if and only if there exists a continuous function  $f : X \rightarrow [0, 1]$  such that  $E = f^{-1}(\{0\})$  and  $F = f^{-1}(\{1\})$ .

A topological space is called a perfectly normal space if and only if any two disjoint closed sets in it are precisely separated by a (continuous) function. A topological space is a perfect normal space implies it is a completely normal space.

### **xix Perfectly normal Hausdorff space, $T_6$ space, or perfectly $T_4$ space**

A topological space is called a perfectly normal Hausdorff space,  $T_6$  space, or perfectly  $T_4$  space if and only if it is a perfectly normal space and a  $T_1$  space. A topological space is a perfectly normal Hausdorff space implies it is a completely normal Hausdorff space.

### **xx Connected space (連通空間)**

A topological space is called disconnected if and only if it can be expressed as the union of two disjoint non-empty open sets; otherwise, it is called connected.

### **xxi Path-connected space (路徑連通空間)**

A topological space  $X$  is called path-connected if and only if for any two points  $x, y \in X$ , there exists a continuous function  $f : [0, 1] \rightarrow X$  such that  $f(0) = x$  and  $f(1) = y$ . A topological space is path-connected implies it is connected.

### **xxii Simply connected space or 1-connected space (單連通空間)**

A topological space is called simply connected if and only if it is path-connected, and that for any two points  $x, y \in X$  and any two continuous functions  $f : [0, 1] \rightarrow X$  and  $g : [0, 1] \rightarrow X$  such that  $f(0) = g(0) = x$  and  $f(1) = g(1) = y$ , there exists a continuous function  $F : [0, 1] \times [0, 1] \rightarrow X$  such that  $F(x, 0) = f(x)$  and  $F(x, 1) = g(x)$ .

### **xxiii Compact (緊緻) (subset)**

A subset  $K$  of a topological space  $X$  is said to be compact if for every collection  $C$  of open subsets of  $X$  such that

$$K \subseteq \bigcup_{S \in C} S,$$

called a open cover, there is a finite subcollection  $F \subseteq C$  such that

$$K \subseteq \bigcup_{S \in F} S,$$

called a finite subcover.

## **xxiv Compact space**

A topological space  $X$  is called compact iff for every collection  $C$  of open subsets of  $X$  such that

$$X = \bigcup_{S \in C} S,$$

called a open cover, there is a finite subcollection  $F \subseteq C$  such that

$$X = \bigcup_{S \in F} S,$$

called a finite subcover.

## **xxv (Weakly) locally compact space**

A topological space  $X$  is called (weakly) locally compact iff every point in it has a compact neighbourhood.

## **xxvi Strongly locally compact or locally relatively compact space**

A topological space  $X$  is called strongly locally compact or locally relatively compact iff every point in it has a closed compact neighbourhood.

A Hausdorff space is weakly locally compact iff it is strongly locally compact.

## **xxvii $\sigma$ -compact space**

A topological space  $X$  is called  $\sigma$ -compact iff its topology is the union of countably many compact spaces.

## **xxviii $\sigma$ -locally compact space**

A topological space  $X$  is called  $\sigma$ -locally compact iff it is  $\sigma$ -compact and locally compact.

## **xxix Second-countable space or completely separable space**

A topological space is second-countable iff it has a countable basis.

## **XVIII Locally finite cover**

Let  $X$  be a topological space, and  $\{U_i \mid i \in I\}$  be a cover of  $X$ . We say that  $U$  is locally finite if for all  $x \in X$ , there exists  $V \ni x$  that is open in  $X$  such that the set  $\{i \in I \mid V \cap U_i \neq \emptyset\}$  is finite.

## **XIX Dense (稠密) subset**

A subset  $A$  of a topological space  $X$  is said to be a dense subset of  $X$  if for every  $x \in X$  and every neighborhood  $U$  of  $x$ ,  $U \cap A \neq \emptyset$ .

## XX Comparison of topologies

Let  $\tau_1$  and  $\tau_2$  be two topologies on a set  $X$ .

$\tau_1$  is said to be a coarser (weaker or smaller) topology than  $\tau_2$ , and  $\tau_2$  is said to be a finer (stronger or larger) topology than  $\tau_1$ , iff  $\tau_1$  is a subset of  $\tau_2$ .

$\tau_1$  is said to be a strictly coarser topology than  $\tau_2$ , and  $\tau_2$  is said to be a strictly finer topology than  $\tau_1$ , iff  $\tau_1$  is a proper subset of  $\tau_2$ .

The binary relation  $\subseteq$  defines a partial ordering relation on the set of all possible topologies on  $X$ .

## XXI Product topology or Tychonoff topology

The product topology or Tychonoff topology on a set  $X = \prod_{i=1}^n X_i$  where  $X_i$  are topological spaces and  $n > 1$  is defined as the coarsest topology on  $X$  such that all  $i$ th projection map are continuous.  $X$  with the product topology on  $X$  is called a product space. Unless otherwise specified, product topology is assumed on a Cartesian product of topological spaces.

## XXII Metric space (度量空間 or 賦距空間)

### i Metric space

Metric space is an ordered pair  $(M, d)$  where  $M$  is a set and  $d$  is a metric on  $M$ , i.e., a function  $d : M \times M \rightarrow \mathbb{R}$  satisfying the following axioms for all points  $x, y, z \in M$ :

1. The distance from a point to itself is zero:  $d(x, x) = 0$ .
2. (Positivity) The distance between two distinct points is always positive:  $x \neq y \implies d(x, y) > 0$ .
3. (Symmetry) The distance from  $x$  to  $y$  is always the same as the distance from  $y$  to  $x$ :  $d(x, y) = d(y, x)$ .
4. The triangle inequality:  $d(x, z) \leq d(x, y) + d(y, z)$ .

For a normed space  $(X, \|\cdot\|)$ , unless otherwise specified, it is equipped with a metric  $d$  defined as

$$d(x, y) = \|x - y\|, \quad \forall x, y \in X.$$

### ii Cauchy sequence (柯西序列)

Given a metric space  $(X, d)$ , a sequence of elements of  $X$ :

$$x_1, x_2, x_3, \dots$$

is Cauchy if for every positive real number  $\varepsilon$  there is a positive integer  $N$  such that for all positive integers  $m, n > N$ :

$$d(x_m, x_n) < \varepsilon.$$

### iii Complete metric space (完備度量空間) or Cauchy space (柯西空間)

A metric space  $(X, d)$  is complete if every Cauchy sequence of points in  $X$  has a limit that is also in  $X$ .

### iv Complete subset or Cauchy subset

A subset  $S$  of a metric space  $(X, d)$  is complete if every Cauchy sequence of points in  $S$  has a limit that is also in  $S$ .

### v (Open) ball (開球)

In a metric space  $(X, d)$ , the open ball or ball of radius  $r > 0$  and centered at the point  $a \in X$ , denoted as  $B_{<r}(a)$ ,  $B_r(a)$ , or  $B(a, r)$ , is defined as:

$$B_r(a) := \{p \in X : d(a, p) < r\}.$$

### vi Closed ball (閉球)

In a metric space  $(X, d)$ , the closed ball of radius  $r \geq 0$  and centered at the point  $a \in X$ , denoted as  $B_{\leq r}(a)$ , is defined as:

$$B_{\leq r}(a) := \{p \in X : d(a, p) \leq r\}.$$

### vii Bounded (有界) subset

A subset  $S$  of a metric space  $(M, d)$  is called bounded if and only if there exists  $r > 0$  such that for all  $s, t \in S$ , we have  $d(s, t) < r$ .

### viii Totally bounded or precompact subset

A subset  $S$  of a metric space  $(M, d)$  is called totally bounded or precompact if and only if there exists  $r > 0$  such that there exists a finite collection of open balls of radius  $r$  that is a cover of  $S$ .

### ix $\varepsilon$ -net

A subset  $E$  of a metric space  $X$  is an  $\varepsilon$ -net for another subset  $A$  of  $X$  if

$$A \subseteq \bigcup_{x \in E} B(x, \varepsilon).$$

If a finite  $\varepsilon$ -net exists for every  $\varepsilon > 0$ , then  $A$  is totally bounded.

### x A compact subset of a Hausdorff space is closed

*Statement:* If a subset  $K$  of a Hausdorff space  $X$  is compact, then it must be closed.

*Proof.* We want to show that  $K$  is closed, that is,  $X \setminus K$  is open.

For any  $a \in X \setminus K$ , for each  $x \in K$ , by the definition of Hausdorff space, there exist a disjoint open set  $U_x$  such that  $x \in U_x$ ,  $a \in V_x$ , and  $U_x \cap V_x = \emptyset$ .

The collection of  $C = \{U_x \mid x \in K\}$  is an open cover of  $K$ . Since  $K$  is compact, a finite subset  $D$  of  $C$  is a cover of  $K$ .

Corresponding to each  $U_i \in D$ , we have an open neighborhood  $V_i$  of  $a$  such that  $U_i \cap V_i = \emptyset$ . Let  $V = \bigcap_{U_i \in D} V_i$ . Then  $V$  is an open neighborhood of  $a$  and is disjoint from  $\bigcup_{U_i \in D} U_i \supseteq K$ , so

$$V \cap K = \emptyset.$$

For every  $a \in X \setminus K$ , we found an open neighborhood  $V$  of  $a$  such that

$$V \subseteq X \setminus K \wedge V \cap K = \emptyset,$$

so  $K$  is closed. □

### **xi A closed subset of a compact set is compact**

*Statement:* If  $K$  is a compact subset of a topological space  $X$  and  $F \subseteq K$  is closed, then  $F$  is compact.

*Proof.* Let  $C$  be an open cover of  $F$ . Then

$$C \cup \{X \setminus F\}$$

is an open cover of  $K$ .

Since  $K$  is compact, there is a finite subcollection  $D = C' \cup \{X \setminus F\}$  of  $C \cup \{X \setminus F\}$  where  $C' \subseteq C$  that is a cover of  $K$ . Then  $C'$  is a finite subcollection of  $C$  that is a cover of  $F$ . Thus  $F$  is compact. □

### **xii A compact subset of a metric space is bounded**

*Statement:* If  $K$  is a compact subset of a metric space  $(X, d)$ , then  $K$  is bounded.

*Proof.* Since  $K$  is compact, for any point  $p \in X$ , there exist a finite collection of open balls

$$\{B(p, n) \mid n \in \mathbb{N} \wedge n \leq m \in \mathbb{N}\}$$

that is a cover of  $K$ , then  $K \subseteq B(p, m)$ , so  $K$  is bounded. □

### **xiii A complete subset of a complete metric space is closed**

*Statement:* If a subset  $A$  of a complete metric space is compact, then  $A$  is closed.

*Proof.* Since a metric space is necessarily a Hausdorff space, any convergent sequence must have a unique limit. Since  $A$  is complete, every Cauchy sequence in  $A$  converges to a point in  $A$ . Thus,  $A$  is closed. □

### **xiv A closed subset of a complete metric space is complete**

*Statement:* If a subset  $A$  of a complete metric space  $X$  is closed, then  $A$  is compact.

*Proof.* Let  $(x_n)$  be a Cauchy sequence in  $A$ . Since the space is complete, it converges to some  $x \in X$ . Since  $A$  is closed in  $X$ , any limit of a convergent sequence in  $A$  must be in  $A$ . Thus,  $A$  is complete. □

## xv A closed and bounded subset of an Euclidean space is compact

*Statement:* If a subset  $A$  of an Euclidean space  $\mathbb{R}^n$  is closed and bounded, then  $A$  is compact.

*Proof.* Let  $(x_k)$  be any sequence in  $A$ . We want to show that it has a convergent subsequence with its limit in  $A$ .

Since  $A$  is bounded, there exists  $M > 0$  such that

$$\|x_k\| \leq M, \quad \forall k.$$

So

$$x_k \in B(0, M), \quad \forall k.$$

By the Bolzano–Weierstrass theorem, there exists a subsequence  $(x_{k_j})$  that converges to some point  $x \in \mathbb{R}^n$ .

Since  $A$  is closed, it is complete. Thus  $x \in A$ . □

## xvi Heine–Borel Theorem (海涅-博雷爾定理)

For a subset  $S$  of an Euclidean space  $\mathbb{R}^n$ , the following two statements are equivalent:

- $S$  is compact;
- $S$  is closed and bounded.

## xvii A complete and totally bounded subset of a complete metric space is compact

*Statement:* If a subset  $A$  of a complete metric space  $(X, d)$  is complete and totally bounded, then  $A$  is compact.

*Proof.* Take any sequence  $(x_n)$  in  $A$ . We want to find a convergent subsequence whose limit lies in  $A$ .

For  $\varepsilon = 1$ , since  $A$  is totally bounded, there exists a finite cover of  $A$  by closed balls of radius 1:

$$A \subseteq \bigcup_{i=1}^{N_1} B_1(y_i).$$

Since  $(x_n)$  is infinite but the cover is finite, by the pigeonhole principle, one of these balls must contain infinitely many terms of  $(x_n)$ . Denote this ball by  $B_1(y_{i_1})$  and pick the subsequence  $(x_n^{(1)})$  of  $(x_n)$  that is entirely in  $B_1(y_{i_1})$ .

Repeat it for  $\varepsilon = \frac{1}{2}, \frac{1}{3}, \dots$ , we can construct a nested sequence of subsequences

$$(x_n^{(1)}) \supseteq (x_n^{(2)}) \supseteq \dots$$

where  $(x_n^{(k)})$  lies in a closed ball of radius  $\frac{1}{k}$ ,  $B_{\frac{1}{k}}(y_{i_k})$ .

Pick the diagonal sequence  $y_k = x_k^{(k)}$ . Then

$$y_k \in (x_n^{(k)}) \subseteq B_{\frac{1}{k}}(y_{i_k}),$$



and the sequence  $(y_k)$  is a Cauchy sequence in  $A$  because

$$\forall N \in \mathbb{N} : m, n \geq N \implies d(y_m, y_n) \leq \frac{2}{N}.$$

Since  $A$  is complete,  $(y_k)$  must converges to some point  $y \in A$ . □

### **xviii Generalized/general Heine–Borel Theorem**

For a subset  $S$  of a complete metric space  $(X, d)$ , the following two statements are equivalent:

- $S$  is compact;
- $S$  is complete and totally bounded.

### **xix A totally bounded subset is bounded**

*Statement:* If a subset  $A$  of a metric space  $(X, d)$  is totally bounded, then  $A$  is bounded.

*Proof.* By total boundedness, there exist finitely many points  $x_1, x_2, \dots, x_n \in X$  such that:

$$A \subseteq \bigcup_{i=1}^n B(x_i, 1).$$

Pick one of these centers, say  $x_1$ .

For any  $a \in A$ , there exists some  $i \in \{1, 2, \dots, n\}$  such that

$$d(a, x_i) < 1.$$

By triangle inequality:

$$d(a, x_1) \leq d(a, x_i) + d(x_i, x_1) \leq 1 + d(x_i, x_1).$$

Let:

$$M = 1 + \max_{i \in \{1, 2, \dots, n\}} d(x_i, x_1).$$

So, for all  $a \in A$ :

$$d(a, x_1) < M.$$

Thus:

$$A \subseteq B(x_1, M).$$

Therefore,  $A$  is bounded. □

### **xx All norms on a finite-dimensional vector space are equivalent**

*Statement:* Let  $V$  be a finite-dimensional vector space over  $\mathbb{R}$  or  $\mathbb{C}$  and  $\dim V = n$ . If  $\|\cdot\|_a$  and  $\|\cdot\|_b$  are two norm on  $V$ , then there exist constants  $c, C > 0$  such that for all  $v \in V$ ,

$$c\|v\|_a \leq \|v\|_b \leq C\|v\|_a,$$

called that the two norms are equivalent, and  $c, C$  are called the equivalence constants.

*Proof.* First consider  $V = \mathbb{R}^n$ . The set

$$S_a = \{x \in \mathbb{R}^n \mid \|x\|_a = 1\}$$

is closed and bounded in  $\mathbb{R}^n$ , hence compact.

Define

$$f(x) = \|x\|_b, \quad x \in S_a.$$

Since  $\|\cdot\|_b$  is continuous,  $f$  is continuous on the compact set  $S_a$ . Therefore  $f$  attains a minimum and maximum on  $S_a$ . So there exist  $m, M \geq 0$  such that

$$m \leq f(x) \leq M.$$

For any  $v \neq 0$ , write

$$u = \frac{v}{\|v\|_a} \in S_a.$$

Then

$$\|v\|_b = \|v\|_a \cdot \|u\|_b.$$

So

$$m\|v\|_a \leq \|v\|_b \leq M\|v\|_a.$$

Thus  $c = m, C = M$  are the equivalence constants. □

## **xxi A bounded subset of a finite-dimensional Banach space is totally bounded**

*Statement:* If a subset  $A$  of a  $n$ -dimensional Banach space  $X$  with  $n \in \mathbb{N}$  is bounded, then  $A$  is totally bounded.

*Proof.* Pick a linear homeomorphism

$$T : \mathbb{R}^n \rightarrow X.$$

Define a norm on  $\mathbb{R}^n$  by

$$\|v\|_* := \|Tv\|_X, \quad \forall v \in \mathbb{R}^n.$$

By that all norms on a finite-dimensional vector space are equivalent, the Euclidean norm  $\|\cdot\|_2$  and  $\|\cdot\|_*$  are equivalent, that is, there exist constants  $c, C > 0$  such that for every  $v \in \mathbb{R}^n$ ,

$$c\|v\|_2 \leq \|v\|_* \leq C\|v\|_2. \quad (1)$$

Let  $A \subseteq X$  be bounded. Then  $B := T^{-1}(A)$  is bounded with respect to the metric induced by  $\|\cdot\|_*$  and, by (1), is bounded with respect to the metric induced by  $\|\cdot\|_2$ . So there is  $R > 0$  with  $B_{\|\cdot\|_2}(0, R) \supseteq B$ . For any  $\varepsilon > 0$  and  $\delta := \frac{\varepsilon}{C}$ , there is a finite collection of  $B_{\|\cdot\|_2}(v_i, R)$  where  $v_i \in \mathbb{R}^n$  that is a cover of  $B_{\|\cdot\|_2}(0, R)$ .

Take any  $b \in B$ . Then  $b \in B_{\|\cdot\|_2}(v_i, R)$  for some  $v_i$ , so  $\|b - v_i\|_2 < \delta$ .

By (1), we get

$$\|T(b) - T(v_i)\|_X = \|b - v_i\|_* \leq C\|b - v_i\|_2 < C\delta = \varepsilon.$$

Therefore the finite set of all  $T(v_i)$  is an  $\varepsilon$ -net for  $A$ . Since  $\varepsilon > 0$  was arbitrary,  $A$  is totally bounded. □

## XXIII Net (網)

### i Net (網)

A net in  $X$ , denoted as  $x_\bullet = (x_a)_{a \in A}$  or  $\langle x_a \rangle_{a \in A}$ , is a function of the form  $x_\bullet: A \rightarrow X$  whose domain  $A$  is some directed set, and whose values are  $x_\bullet(a) = x_a$  in a topological space  $X$ . Elements of a net's domain are called its indices.

### ii Eventually or residually

A net  $x_\bullet = (x_a)_{a \in A}$  in  $X$  is said to be eventually or residually in a set  $S$  if there exists some  $a \in A$  such that for every  $b \in A$  with  $b \geq a$ , the point  $x_b \in S$ .

### iii Limit of nets

A point  $x$  in a topological space  $X$  is called a limit point or limit of a net  $x_\bullet$  in  $X$  if, for every open neighborhood  $U$  of  $x$ , the net  $x_\bullet$  is eventually in  $U$ , also called that the net converges to/towards  $x$ , and denoted as  $x_\bullet \rightarrow x$ ,  $x_a \rightarrow x$ ,  $\lim x_\bullet \rightarrow x$ ,  $\lim_{a \in A} x_a \rightarrow x$ , or  $\lim_a x_a \rightarrow x$ .

If  $\lim x_\bullet \rightarrow x$  and this limit is unique (i.e.  $\lim x_\bullet \rightarrow y$  only if  $x = y$ ), then one writes  $\lim x_\bullet = x$ ,  $\lim_{a \in A} x_a = x$ , or  $\lim_a x_a = x$ .

### iv Cofinal (共尾) or frequent subset

A subset  $B$  of a directed set  $A$  is a cofinal or frequent subset if for all  $a \in A$ , there exists  $b \in B$  such that  $b \geq a$ .

### v Cofinally (共尾地) or frequently

A net  $x_\bullet = (x_a)_{a \in A}$  in  $X$  is said to be cofinally or frequently in a set  $S$  if for every  $a \in A$  there exists  $b \in A$  such that  $b \geq a \wedge x_b \in S$ .

### vi Cluster point, or accumulation point (集積點)

A point  $x$  in a topological space  $X$  is called a cluster point or accumulation point of a net  $x_\bullet$  in  $X$  if, for every open neighborhood  $U$  of  $x$ , the net  $x_\bullet$  is cofinally in  $U$ , also called that the net clusters to/towards  $x$ .

A limit point of a net is necessary a cluster point of it.

### vii Subnet or Willard-subnet

A net  $s_\bullet = (s_i)_{i \in I}$  is called a subnet or Willard-subnet of another net  $x_\bullet = (x_a)_{a \in A}$  if there exists an order-preserving map  $h: I \rightarrow A$  such that  $h(I)$  is a cofinal subset of  $A$  and

$$s_i = x_{h(i)} \quad \forall i \in I.$$

## viii Ultranet or universal net

A net  $x_\bullet$  in  $X$  is called an ultranet or a universal net if for every subset  $S \subseteq X$ ,  $x_\bullet$  is eventually in  $S$  or  $X \setminus S$ .

A point is a limit point of a ultranet if and only if it a cluster point of it.

## ix In Hausdorff spaces

A net in a Hausdorff space has at most one limit; if it has one, the limit point of it is the only cluster point of it.

*Proof.* TODO □

## x Willard-subnet limit and cluster point theorems

For any nonempty net  $\langle a_n \rangle_{n \in A}$  and its any Willard-subnet  $\langle b_n \rangle_{n \in B}$ , if  $x$  is a limit point of  $a_n$ , then  $x$  is a limit point of  $b_n$ .

*Proof.* Let  $h: B \rightarrow A$  be order-preserving and  $h(B)$  be a cofinal subset of  $A$ . Let  $U$  be an open neighborhood of  $x$ . There exists  $a \in A$  such that  $n \geq a \implies a_n \in U$ . Then, for any  $b \in B$  such that  $h(b) \geq a$ ,  $m \geq b \implies b_m = a_{h(m)} \in U$ . □

For any nonempty net  $\langle a_n \rangle_{n \in A}$  and its any Willard-subnet  $\langle b_n \rangle_{n \in B}$ , if  $x$  is a cluster point of  $b_n$ , then  $x$  is a cluster point of  $a_n$ .

*Proof.* Let  $h: B \rightarrow A$  be order-preserving and  $h(B)$  be a cofinal subset of  $A$ . Let  $U$  be an open neighborhood of  $x$ . For any  $b \in B$ , there exists  $m \geq b$  such that  $a_{h(m)} = b_m \in U$ . Also, for any  $c \in A$ , there exists  $d \in B$  such that  $h(d) \geq c$ . Thus, for any  $a \in A$ , there exists  $n \geq a$  such that  $a_n \in U$ . □

Its special case when  $A = B = \mathbb{N}_0$  or  $\mathbb{N}_1$ , and  $a_n \in \mathbb{R}$  is called the subsequence convergence theorem.

# XXIV Manifold

## i Euclidean neighborhood

An Euclidean neighborhood of a point in a topological space is a neighborhood of it that is homeomorphic to an open subset of  $\mathbb{R}^n$  for some  $n \in \mathbb{N}_0$ . Such Euclidean neighborhood is called  $n$ -dimensional.

A Euclidean neighborhood homeomorphic to an open ball in  $\mathbb{R}^n$  is called a Euclidean ball.

## ii Locally Euclidean space

A topological space  $X$  is called locally Euclidean iff it can be covered by  $n$ -dimensional Euclidean neighborhoods for some  $n \in \mathbb{N}_0$ . Such locally Euclidean space is called  $n$ -dimensional or an  $n$ -locally Euclidean space.

Euclidean balls form a basis for the topology of a locally Euclidean space.

### iii (Topological) manifold (流形)

A (topological) manifold is a locally Euclidean space that is a Hausdorff space. An  $n$ -dimensional (topological) manifold or  $n$ -manifold is an  $n$ -dimensional locally Euclidean space that is a Hausdorff space.

An 1D manifold is called curve. A 2D manifold is called a surface.

### iv (Coordinate) chart

For any Euclidean neighborhood  $U$ , a homeomorphism  $\phi: U \rightarrow \phi(U) \subseteq \mathbb{R}^n$  is called a (coordinate) chart on  $U$ .

### v Atlas

For a locally Euclidean space  $M$ , an atlas  $\{(U_i, \phi_i) \mid i \in I\}$  on  $M$  is a family of ordered pairs of Euclidean neighborhood  $U_i$  and coordinate chart  $\phi_i$  on  $U_i$  such that  $\{U_i \mid i \in I\}$  covers  $M$ .

### vi Transition function or transition map

For an atlas  $\{(U_i, \phi_i) \mid i \in I\}$  on an  $n$ -dimensional locally Euclidean space  $M$ , the transition function  $\phi_i \phi_j^{-1}$  for  $i, j \in I$  is defined as the function

$$\phi_i \phi_j^{-1} : \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j); x \mapsto \phi_i(\phi_j^{-1}(x)).$$

Such a map is a homeomorphism between open subsets of  $\mathbb{R}^n$ .

### vii Differentiable and smooth

An atlas  $\{(U_i, \phi_i) \mid i \in I\}$  is called a differentiable atlas iff for all  $i, j \in I$ , the transition map  $\phi_i \phi_j^{-1}$  is differentiable.

An atlas  $\{(U_i, \phi_i) \mid i \in I\}$  is called a ( $n$ -times) continuously differentiable atlas iff for all  $i, j \in I$ , the transition map  $\phi_i \phi_j^{-1}$  is ( $n$ -times) continuously differentiable.

An atlas  $\{(U_i, \phi_i) \mid i \in I\}$  is called a  $C^k$  atlas iff for all  $i, j \in I$ , the transition map  $\phi_i \phi_j^{-1}$  is a  $C^k$  function.

An atlas  $\{(U_i, \phi_i) \mid i \in I\}$  is called a smooth ( $C^\infty$ ) atlas iff for all  $i, j \in I$ , the transition map  $\phi_i \phi_j^{-1}$  is a smooth ( $C^\infty$ ) function.

A differentiable manifold is a manifold such that there exists a differentiable atlas on it.

A ( $n$ -times) continuously differentiable manifold is a manifold such that there exists a ( $n$ -times) continuously differentiable atlas on it.

A  $C^k$  manifold is a manifold such that there exists a  $C^k$  atlas on it.

A smooth ( $C^\infty$ ) manifold is a manifold such that there exists a smooth ( $C^\infty$ ) atlas on it.

Let  $M$  be an  $n$ -dimensional differentiable manifold with differentiable atlas  $\{(U_i, \phi_i) \mid i \in I\}$ . A map  $f : M \rightarrow M$  is a differentiable map iff for any charts  $(U_i, \phi_i)$  such that  $p \in U_i \wedge i \in I$  and  $(U_j, \phi_j)$  such that  $f(p) \in U_j \wedge j \in I$ ,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$$

is differentiable.

Let  $M$  be an  $n$ -dimensional ( $n$ -times) continuously differentiable manifold with ( $n$ -times) continuously differentiable atlas  $\{(U_i, \phi_i) \mid i \in I\}$ . A map  $f : M \rightarrow M$  is an ( $n$ -times) continuously differentiable map iff for any charts  $(U_i, \phi_i)$  such that  $p \in U_i \wedge i \in I$  and  $(U_j, \phi_j)$  such that  $f(p) \in U_j \wedge j \in I$ ,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$$

is ( $n$ -times) continuously differentiable.

Let  $M$  be an  $n$ -dimensional  $C^k$  manifold with  $C^k$  atlas  $\{(U_i, \phi_i) \mid i \in I\}$ . A map  $f : M \rightarrow M$  is a  $C^k$  map iff for any charts  $(U_i, \phi_i)$  such that  $p \in U_i \wedge i \in I$  and  $(U_j, \phi_j)$  such that  $f(p) \in U_j \wedge j \in I$ ,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$$

is a  $C^k$  function.

Let  $M$  be an  $n$ -dimensional smooth ( $C^\infty$ ) manifold with smooth ( $C^\infty$ ) atlas  $\{(U_i, \phi_i) \mid i \in I\}$ . A map  $f : M \rightarrow M$  is a smooth ( $C^\infty$ ) map iff for any charts  $(U_i, \phi_i)$  such that  $p \in U_i \wedge i \in I$  and  $(U_j, \phi_j)$  such that  $f(p) \in U_j \wedge j \in I$ ,

$$\phi_j \circ f \circ \phi_i^{-1} : \phi_i(U_i \cap f^{-1}(U_j)) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$$

is a smooth ( $C^\infty$ ) function.

### viii Diffeomorphism

Given two differentiable manifolds  $X, Y$ , a continuously differentiable map  $f : X \rightarrow Y$  is a diffeomorphism iff it is a bijection and its inverse is differentiable. If there exists a diffeomorphism between two differentiable manifolds, they are called diffeomorphic.

Given two  $C^k$  manifolds  $X, Y$ , a  $C^k$  map  $f : X \rightarrow Y$  is a  $C^k$ -diffeomorphism iff it is a bijection and its inverse is  $C^k$ . If there exists a  $C^k$ -diffeomorphism between two  $C^k$  manifolds, they are called  $C^k$ -diffeomorphic.

### ix Local diffeomorphism

Given two differentiable manifolds  $X, Y$ , a map  $f : X \rightarrow Y$  is a local diffeomorphism iff for any point  $x \in X$ , there exists an open neighborhood  $U$  of  $x$  such that  $f_U : U \rightarrow f(U)$  is a diffeomorphism. If there exists a local diffeomorphism between two differentiable manifolds, they are called locally diffeomorphic.

Given two  $C^k$  manifolds  $X, Y$ , a map  $f : X \rightarrow Y$  is a local  $C^k$ -diffeomorphism iff for any point  $x \in X$ , there exists an open neighborhood  $U$  of  $x$  such that  $f_U : U \rightarrow f(U)$  is a  $C^k$ -diffeomorphism. If there exists a local  $C^k$ -diffeomorphism between two differentiable manifolds, they are called locally  $C^k$ -diffeomorphic.

## **XXV Combination with abstract algebra**

### **i Topological group (拓撲群)**

A topological group is a triple  $(X, \tau, *)$  such that  $(X, \tau)$  is a topological space,  $(X, *)$  is a group, and the maps of  $(x, y) \in X^2 \mapsto x * y$  and  $x \mapsto x^{-1}$  are continuous maps. Such topology  $\tau$  is called a group topology.

### **ii Topological field (拓撲域)**

A topological field is defined with a topological space  $(X, \tau)$  and a field  $(X, +, *)$  such that the maps of  $a \mapsto -a$  and  $a \mapsto a^{-1}$  are continuous maps.

## **XXVI Discrete space**

### **i Discrete topology**

A discrete topology on  $X$  is a topology such that every subset of  $X$  is open.

### **ii Discrete (topological) space**

A space equipped with discrete topology.

### **iii Discrete subspace**

A topological subspace whose subspace topology is a discrete topology.